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|                 | $T_A = 280 \text{ K}$                                                                                                                                                                                                                                                                                                                                                                      |
| <b>7(a)</b>     | The electric field strength at a point in an electric field is defined as the electric force exerted per unit positive charge placed at that point.                                                                                                                                                                                                                                        |
| <b>7(b)(i)</b>  |  <p>A diagram showing a central grey circle with a minus sign inside, representing a negative point charge. Eight straight lines with arrows point radially inward towards the circle, representing the electric field lines of a negative charge.</p>                                                    |
| <b>7(b)(ii)</b> | <p>The electric field strength at a point is the negative of the potential gradient at that point. Thus, the magnitude of the electric field strength at a distance of 40 cm from the surface of the sphere is <math>130 \text{ V m}^{-1}</math>.</p> $E = \frac{Q}{4\pi\epsilon_0 r^2}$ $130 = \frac{Q}{4\pi (8.85 \times 10^{-12}) (0.15 + 0.40)^2}$ $Q = 4.37 \times 10^{-9} \text{ C}$ |
| <b>8(a)(i)</b>  | The spring constant can be found from the gradient of the force extension graph                                                                                                                                                                                                                                                                                                            |